

Algorithmic Multi-Asset Portfolio Optimization Framework

Lead Researcher & Author: Busra Akkaya

Study Type: Independent Quantitative Research & Capital Market Allocation Study

Theorists & Foundations: Markowitz Modern Portfolio Theory (MPT), Capital Asset Pricing Model (CAPM)

Asset Universe: Large-Cap US Equities (*SPY*), Regional UK Equities (*FTAL.L*), Core Fixed Income (*IEGA.L*), Commodities / Safe-Haven Alternatives (*GLD*)

1. Executive Summary & Financial Architecture

This quantitative architecture establishes a production-grade framework for optimizing capital allocation across a diversified multi-asset universe. Rooted in Harry Markowitz's foundational 1952 Modern Portfolio Theory (MPT), the framework moves beyond simple descriptive performance metrics to evaluate assets entirely through their joint risk-covariance structures. By mapping out a stochastic sample space of 10,000 distinct asset configurations, this pipeline algorithmically solves for the multi-asset Efficient Frontier, providing deterministic solutions optimized for risk-adjusted yield or comprehensive capital preservation.

The core framework incorporates Capital Market Theory baselines to assess systematic risk filters (β) in accordance with the Sharpe-Lintner-Treynor model, evaluating portfolios against an institutional risk-free proxy (R_f) baseline of 4.0% to simulate realistic current macroeconomic hurdle rules.

2. Data Ingestion & Cleansing Pipeline

2.1 The Multi-Asset Pipeline

To capture realistic global market variations, the framework institutes a live data acquisition pipeline utilizing Yahoo Finance API endpoints to stream 5 consecutive years of daily closing price time-series. Rather than holding isolated concentrated single equities, the asset space uses highly liquid exchange-traded funds (ETFs) and indices across global macro regions:

- **FTAL.L** (SPDR FTSE UK All Share ETF) — Regional UK Equities factor proxy.
- **GLD** (SPDR Gold Shares) — Alternative safe-haven commodity exposure.
- **IEGA.L** (iShares Euro Government Bond ETF) — Sovereign core fixed-income baseline.
- **SPY** (S&P 500 ETF Trust) — Large-Cap US Equities core growth engine.

2.2 Data Wrangling and Transformation

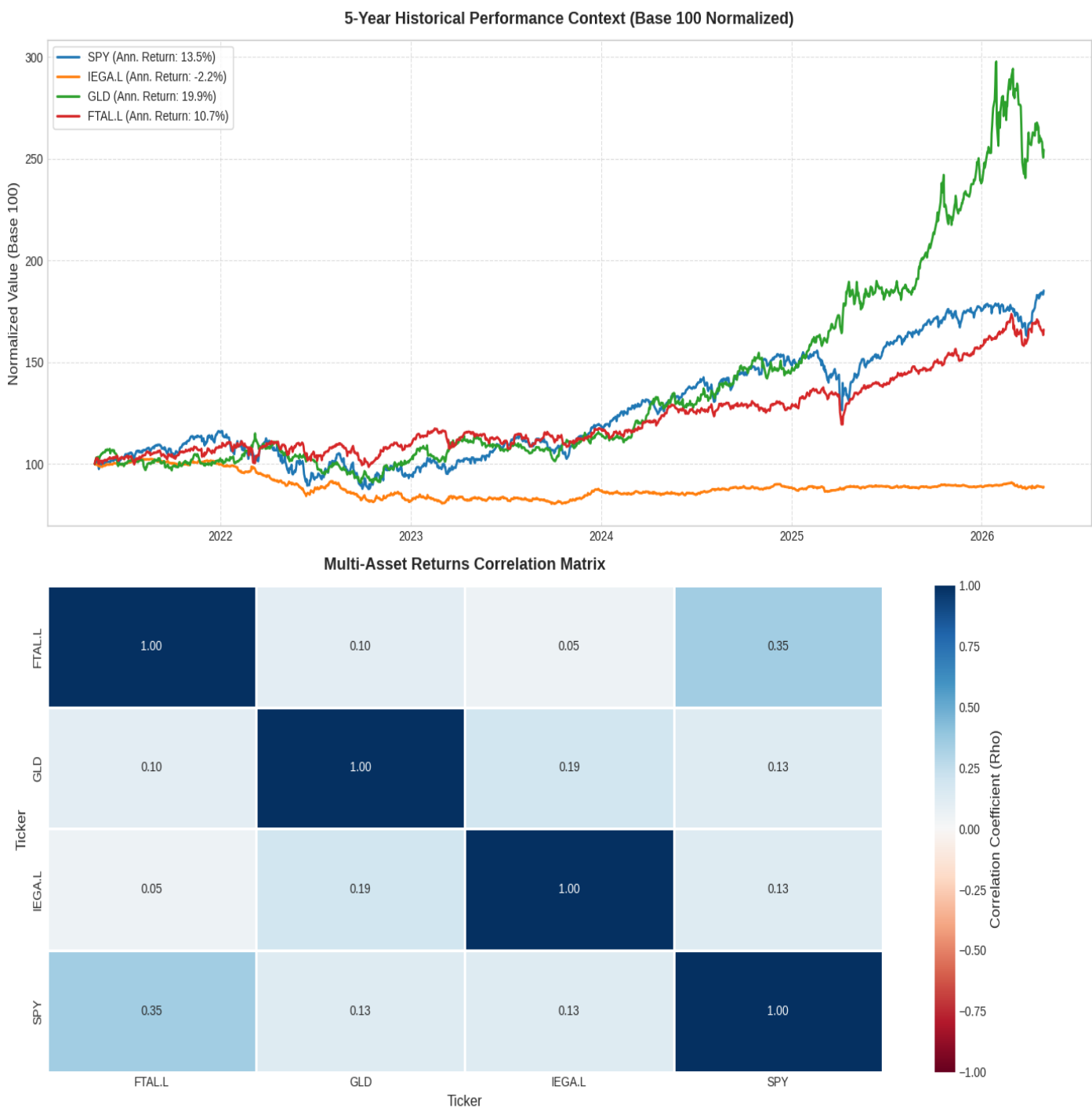
Asynchronous trading calendars across international exchanges (e.g., London Stock Exchange vs. New York Stock Exchange market closure regimes) generate structural missing data cells (*NaN* observations). The data engineering pipeline deploys a strict forward-fill (*ffill*) cleansing methodology to retain data sequence

continuity without embedding looking-ahead or tracking bias. Raw prices are mathematically transformed into stationary daily percentage returns for stable statistical modeling.

3. Exploratory Visual Analytics & Correlation Structure

3.1 Empirical Market Performance (5-Year Cycle)

Historical evaluation confirms highly divergent trends across the asset classes. Gold (*GLD*) served as a massive cyclical outlier, capturing intense upside acceleration to deliver a remarkable **19.9% annualized return**. US Large-Cap Equities (*SPY*) provided robust compound growth at **13.5%**, while UK regional equity fields (*FTAL.L*) secured a moderate **10.7%**. Conversely, Euro Government Bonds (*IEGAL*) suffered from global rate transitions, yielding a negative standalone return of **-2.2%**.



3.2 Cross-Asset Structural Correlation Matrix

The statistical returns matrix exposes the underlying source of mathematical diversification benefits, revealing exceptionally low cross-asset correlation profiles (ρ) sitting near zero:

- **Equities vs. Bonds:** UK Equities (*FTALL*) and Euro Bonds (*IEGAL*) are completely uncorrelated at $\rho = 0.05$. Similarly, US Equities (*SPY*) and Euro Bonds share a minimal connection of $\rho = 0.13$.
- **Gold Factor Independence:** Gold (*GLD*) exhibits excellent alternative characteristics, maintaining a low correlation of $\rho = 0.10$ with UK Equities and $\rho = 0.13$ with US Equities.

This structural arrangement mathematically justifies the deployment of MPT. Because cross-correlations approach zero ($\rho \rightarrow 0$), the combined joint variance of the assets will be materially dampened, allowing the optimization loop to effectively compress total risk profiles.

4. The Monte Carlo Simulation Engine

4.1 Stochastic Portfolio Mapping

To computationally map out the entire risk-return continuous boundary, a high-performance stochastic engine generates **10,000 randomized long-only asset weight configurations**. For each discrete loop iteration, the engine enforces the absolute institutional constraint that the allocation budget must be fully invested:

$$\sum w_i = 1.0 \quad \text{where} \quad w_i \geq 0$$

4.2 Computational Metrics Vectorization

For each iteration, the engine executes optimized matrix dot-product expressions to derive annualized performance coordinates:

- **Expected Return (R_p):** Vector multiplication of the weight array against historical annualized mean returns.

- **Portfolio Volatility (σ_p):** Calculated via the quadratic form utilizing the annualized covariance matrix Σ : $\sigma_p = \sqrt{(w^T \Sigma w)}$
- **Risk-Adjusted Performance (Sharpe Ratio):** Evaluated using the institutional proxy threshold: $SR = (R_p - 0.04) / \sigma_p$

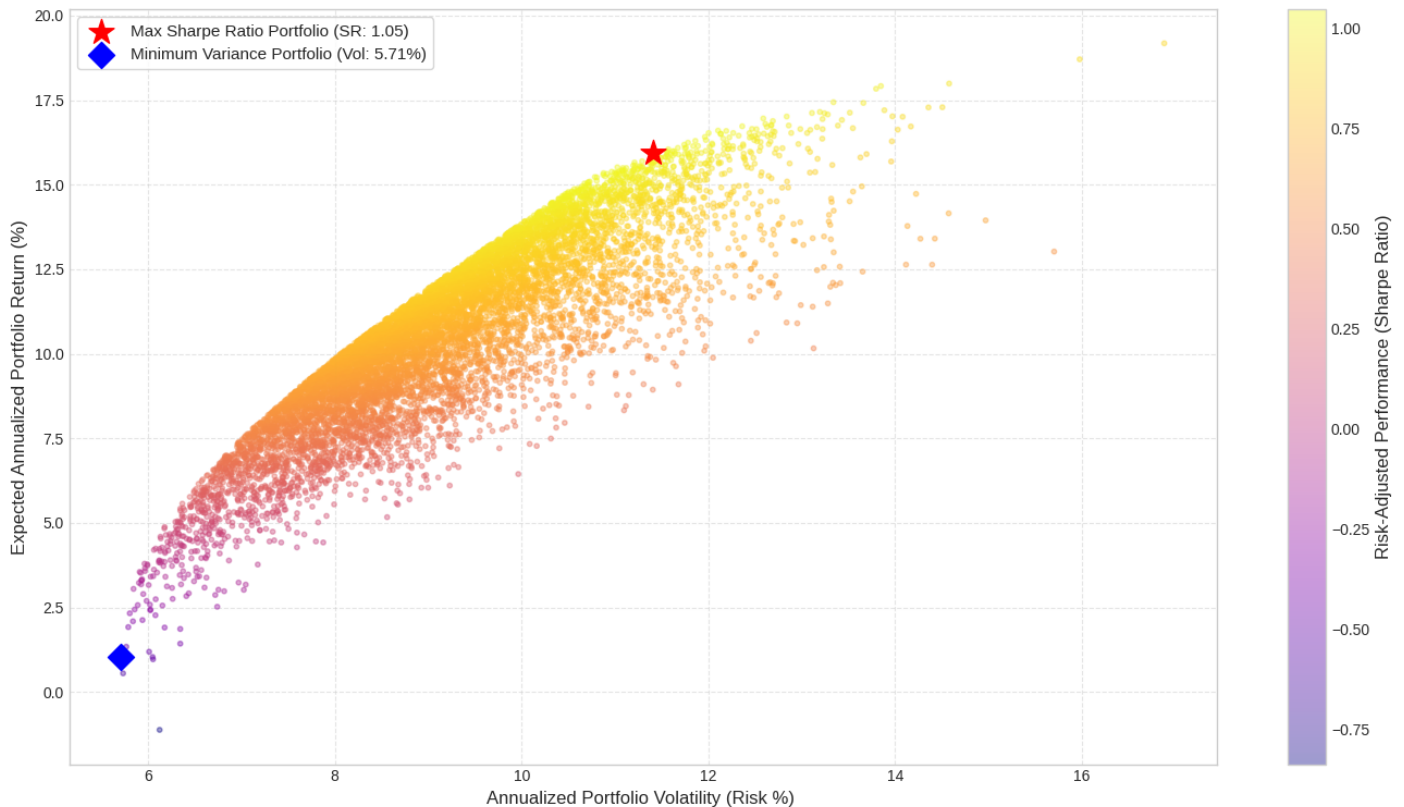
The simulation loop successfully maps a broad risk frontier: a compressed minimum risk floor of **5.71% volatility** up to an exceptional peak Sharpe Ratio of **1.05**, establishing the dataset required for formal optimization indexing.

5. Portfolio Optimization & Frontier Isolate Solutions

Programmatic indexing isolates the two optimal capital structures defined by institutional fund mandates:

Asset / Portfolio Metric	Max Sharpe Ratio (MSR) Red Star	Minimum Variance (Min Var) Blue Diamond
FTAL.L (UK Large-Cap Equities)	30.88%	16.63%
GLD (SPDR Gold Shares)	52.35%	3.42%
IEGA.L (Euro Government Bonds)	0.19%	78.15%
SPY (US Large-Cap Equities)	16.59%	1.79%
Expected Annualized Return	15.96%	1.03%
Total Portfolio Volatility (Risk)	11.41%	5.71%
Portfolio Sharpe Ratio	1.05	-0.52

The Multi-Asset Efficient Frontier Curve (Stochastic Mapping)



5.1 Deep-Dive Strategic Allocation Logic

The Maximum Sharpe Target (1.05): To optimize risk-adjusted performance, the algorithm heavily concentrates capital into the cycle's dominant alpha engine: Gold (*GLD* at 52.35%), reinforced by core growth equity layers. Rather than over-allocating into a single asset, the engine utilizes the low equity-to-gold cross-correlation to capture major return upside while anchoring collective volatility to a modest 11.41% channel.

The Minimum Variance Target (5.71% Risk Floor): Answering an absolute capital preservation mandate, the engine completely rotates capital out of risk assets, allocating a massive **78.15% into Euro Government Bonds (*IEGAL*)**. While bonds generated negative standalone returns over this window, their zero-covariance behavior acts as a structural volatility sponge, driving portfolio risk to the absolute lowest boundary possible within the asset universe.

Technical Conclusion & Production Readiness

This independent framework demonstrates a production-ready data engineering pipeline for front-office asset management environments. By managing asynchronous international multi-asset listings, mapping covariance dimensions, and executing high-speed vector metrics, the system removes qualitative guesswork, replacing it with deterministic allocation boundaries on the Efficient Frontier. The resulting structured time-series and stationary return vectors provide a scalable foundation ready to connect directly into advanced institutional forecasting architectures, such as Black-Litterman adjustments or reinforcement learning allocation models.